R E S 752 T RES

Encyclopaedia Britannica (1771)

Historical source

Entry ID: `entry-r-e-s-752-t-res-611`
Volume ID: 191320558
Character Count: 6222

Extracted Text

Radiance curvature will be infinitly, and the fluid included in the frace ZYHQ will move uniformly as if the field body were not there. The motions on the other side of the axis A,C will be the faming and to read only at tend to our half, and we could conclude as its a late of presence.

No body changes either its direction or velocity other wile than by infentible degrees; therefore the particle which moving the axis will not include the vortex A of the body, where it behaved to deficit influence easily at right angles. It will therefore begin to be diseflected at fromen F is head of the body; and will de-ferible a curve FM, touching the axis in F, and the bo-ly in M, and then, yielding along the body, will gain it at some point I, defening a tangent curve, which will join the axis again (noching it) in B; and thus there will be a quantity of fragment water FAM because or a-head of the body, and another LGR behind or sitem of it.

Let us be the velocity of a particle of the fluid in any infant, and d its velocity in the next instant. The velocity may be considered in compounded of d a. If the particles tended to meet with the velocities da", the whole fluid would be in equilibrium (general principle), and the preference of the fluid would be the films as it all were fragment, and each particle were urged by a force dr, t expelling an indenticity final moment of time. (N. B. jt the proper expedition of the accelerating forces, which, by sliding during the mo- ment r, would generate the velocity ds and d is lip- pooled in determining quantities different perhaps for each particle). Now that a he lopped control, or dazd. In this case dz... That is to fix, no profile whatever will be carried on the fold body until there happen changes in the velocities or directions of the par-ticles.

Let o and d then be the motions of the particles in two confine infinite. They would be in equilibrio if urged only by the forces dr. Therefore if y be the point where the particles which define the one: FM begins to change their velocity, the profiles in D would be equal to the profile when the fluid contained in the canal y EMD would exert, if each particle were folicted by its force dr. The question is therefore reduced to the finding the curvature in the canal y EMD, and the accelerating force dr in its different parts.

It appears, in the first phase that to preface it by any of the particles along the cause FM : for flipside that the particle a (fig. 10,) defective the in- definitely limit through line ad at in the full nation, and b c in the forced infants; produce d till bd=ab, and joining ds a motion and dd may be considered as counted at bg, which the particular reality takes in the next instant, and a motion de which should be defrayed. Draw ds parallel to da and perpendicular to it. It is thin that the particle b, foliated by the forces bg, e/s (equivalent to de) should be in equilibrio. This being obulatively, be mull be = ,q, that is, there will be no accelerating or rounding force at k. For if there

be, draw km (fig. 20) perpendicular to bF, and the Reflines parallel eq infinitely mean it. The part bn of the fluid contained in the canal hngm would inflation force prefix. find out towards, or from a recorded. Therefore face the fluid in this figure can cancel should be in equilibrio, them and also le forms sellon, as leaf in one of the part bm, n,e, q or to counterbalance the chin on the past bm. But the fluid is important in the space FAM (in consequence of the law so continuity). Therethere is no force which can add on bm,

w, q, qy, and the prefiner in the canal in the direction bn or w is nothing, or the force bm, and the force te is perpendicular to the men, and there is therefore no prefiner in the canal FM, except what proceeds from part y F from the force e/, which hit being perpendicular to the canal, there can be no force exerted on the point M, but what is propagated from the part y P.

The velocity therefore in the canal FM is constant if finite, or individual fault variables: for, in the first calc, the force bc would be absolutely nothing; and in the second cad, i would be an infinitesimal of the order, and may be conducted as nothing in com- prantion with the velocity, which is of the first order. We fail too long way that the lab is the real take of the cafe. Therefore the fluid, before it begins to change its distance in N, begun to change its velocity in some point y-haet of Y, and by the time that it reaches F its velocity is as it were annihilated.

Or? it. Therefore the perform in any point D swife from the translation in the part y F, and from the particles which are in the canal MD : as their left move along the surface of the body, the force g, deflyerted in every particle, is composed around of two others, one in the direction of the inferior, and the other spec pedicular to it; call thick p and j. The point D is perfect perpendicular to the surface MD, yth, by all the forces j in the curve MD; z; i, by the theory ed- ing on the fingle point D. This may be neglected in compulsion of the indefinite number of the others therefore taking in the such MAB, an infinitely small portion Nw,=i, the prefine on D, perpendicular to the surface of the body, will be = fjf, and this flu- ent mail bo fo taken as to be = din the point M. Cox . Therefore, to find the precise on D, we must find the force j on any point N. Let a be the velocity of the particle N, in the direction Nw in any infant, and a q-a its velocity in the following infant:

we mutl have p - Therefore the whole quotient is reduced to finding the velocity a in every point N, in the direction Nw.

And this is the sum of a factor of proportions which its neat follow, in which the author defines the math accurate equation and predict conception of the difficulty and great candidate(y) forces and elegance in his mathematical analysis. He si length plan, but brings out an motion which expedite the practice on the body in the mold, ground and manageable pun-ner. We cannot give an abstract, because the train of reading is already coincide in the extreme: nor can we even exhibit the final equation) for it is conceivable in the

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Reſiſtance. curvature will be inſenſible, and the fluid included in the ſpace ZYHQ will move uniformly as if the ſolid body were not there. The motions on the other ſide of the axis AC will be the ſame; and we need only attend to one half, and we ſhall conſider theſe as in a ſtate of permanency.

No body changes either its direction or velocity otherwiſe than by inſenſible degrees: therefore the particle which is moving in the axis will not reach the vertex A of the body, where it behoved to deflect inſtantaneouſly at right angles. It will therefore begin to be deflected at ſome point F a-head of the body, and will deſcribe a curve FM, touching the axis in F, and the body in M; and then, gliding along the body, will quit it at ſome point L, deſcribing a tangent curve, which will join the axis again (touching it) in R; and thus there will be a quantity of ſtagnant water FAM before or a-head of the body, and another LCR behind or aſtern of it.

Let a be the velocity of a particle of the fluid in any inſtant, and a' its velocity in the next inſtant. The velocity a may be conſidered as compounded of a' and a''. If the particles tended to move with the velocities a'' only, the whole fluid would be in equilibrio (general principle), and the preſſure of the fluid would be the ſame as if all were ſtagnant, and each particle were urged by a force $\frac{a''}{t}$, t expreſſing an indefinitely ſmall moment of time. (N. B. $\frac{a''}{t}$ is the proper expreſſion of the accelerating force, which, by acting during the moment t, would generate the velocity a''; and a'' is ſuppoſed an indeterminate quantity, different perhaps for each particle). Now let a be ſuppoſed conſtant, or $a=a'$. In this caſe $a''=o$. That is to ſay, no preſſure whatever will be exerted on the ſolid body unleſs there happen changes in the velocities or directions of the particles.

Let a and a' then be the motions of the particles in two conſecutive inſtants. They would be in equilibrio if urged only by the forces $\frac{a''}{t}$. Therefore if γ be the point where the particles which deſcribe the curve FM begin to change their velocity, the preſſure in D would be equal to the preſſure which the fluid contained in the canal γFMD would exert, if each particle were ſolicited by its force $\frac{a''}{t}$. The queſtion is therefore reduced to the finding the curvature in the canal γFMD, and the accelerating forces $\frac{a''}{t}$ in its different parts.

It appears, in the firſt place, that no preſſure is exerted by any of the particles along the curve FM: for
Fig. 19. ſuppoſe that the particle a (fig. 19.) deſcribes the indefinitely ſmall ſtraight line ab in the firſt inſtant, and bc in the ſecond inſtant; produce ab till $bd=ab$, and joining dc, the motion ab or bd may be conſidered as compoſed of bc, which the particle really takes in the next inſtant, and a motion dc which ſhould be deſtroyed. Draw bi parallel to dc, and ie perpendicular to bc. It is plain that the particle b, ſolicited by the forces bc, ei (equivalent to dc) ſhould be in equilibrio. This being eſtabliſhed, be muſt be $=o$, that is, there will be no accelerating or retarding force at b; for if there be, draw bm (fig. 20.) perpendicular to bF, and the Reſiſtance.
parallel nq infinitely near it. The part bn of the fluid Fig. 20.
contained in the canal $bnqm$ would ſuſtain ſome preſſure from b towards n, or from n towards b. Therefore ſince the fluid in this ſtagnant canal ſhould be in equilibrio, there muſt alſo be ſome action, at leaſt in one of the parts bm, mq, qn, to counterbalance the action on the part bn. But the fluid is ſtagnant in the ſpace FAM (in conſequence of the law of continuity). Therefore there is no force which can act on bm, mq, qn; and the preſſure in the canal in the direction bn or nb is nothing, or the force $be=o$, and the force ie is perpendicular to the canal; and there is therefore no preſſure in the canal FM, except what proceeds from the part γF, or from the force ei; which laſt being perpendicular to the canal, there can be no force exerted on the point M, but what is propagated from the part γF.

The velocity therefore in the canal FM is conſtant if finite, or infinitely ſmall if variable: for, in the firſt caſe, the force be would be abſolutely nothing; and in the ſecond caſe, it would be an infinitеſimal of the ſecond order, and may be conſidered as nothing in compariſon with the velocity, which is of the firſt order. We ſhall ſee by and by that the laſt is the real ſtate of the caſe. Therefore the fluid, before it begins to change its direction in F, begins to change its velocity in ſome point γ a-head of F, and by the time that it reaches F its velocity is as it were annihilated.

Cor. 1. Therefore the preſſure in any point D ariſes both from the retardations in the part γF, and from the particles which are in the canal MD: as theſe laſt move along the ſurface of the body, the force $\frac{a''}{t}$, deſtroyed in every particle, is compounded of two others, one in the direction of the ſurface, and the other perpendicular to it; call theſe p and p'. The point D is preſſed perpendicularly to the ſurface MD; 1ſt, by all the forces p in the curve MD; 2d, by the force p' acting on the ſingle point D. This may be neglected in compariſon of the indefinite number of the others: therefore taking in the arch MD, an infinitely ſmall portion Nm, $=s$, the preſſure on D, perpendicular to the ſurface of the body, will be $=\int p\dot{s}$; and this fluent muſt be ſo taken as to be $=o$ in the point M.

Cor. 2. Therefore, to find the preſſure on D, we muſt find the force p on any point N. Let u be the velocity of the particle N, in the direction Nm in any inſtant, and $u+\dot{u}$ its velocity in the following inſtant; we muſt have $p=\frac{-\dot{u}}{t}$. Therefore the whole queſtion is reduced to finding the velocity u in every point N, in the direction Nm.

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And this is the aim of a ſeries of propoſitions which follow, in which the author diſplays the moſt accurate and preciſe conception of the ſubject, and great addreſs and elegance in his mathematical analyſis. He at length brings out an equation which expreſſes the preſſure on the body in the moſt general and unexceptionable manner. We cannot give an abſtract, becauſe the train of reaſoning is already conciſe in the extreme: nor can we even exhibit the final equation; for it is conceived in the

His final equation truly ſolves the problem, but,

Figure 1: *
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